

Advanced (Habanero)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** Two lines have slopes $m_1 = 3$ and $m_2 = -\frac{1}{3}$. What can be concluded about these lines?
(A) Parallel
(B) Perpendicular
(C) Neither
(D) Same line
(E) Inverse
- Q3.** If a line has a slope of -2 , what is the slope of its perpendicular line?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q4.** Which equation represents a line perpendicular to $y = x - 3$?
(A) $y = -x + 2$
(B) $y = -x - 2$
(C) $y = x + 2$
(D) $y = -x + 1$
(E) $y = x - 2$
- Q5.** The line $y = 4x + 2$ has a perpendicular line passing through $(1, 5)$. What is the equation of this line?
(A) $y = -\frac{1}{4}x + \frac{21}{4}$
(B) $y = -\frac{1}{4}x + \frac{9}{4}$
(C) $y = -\frac{1}{4}x + \frac{17}{4}$
(D) $y = -\frac{1}{4}x + \frac{13}{4}$
(E) $y = -\frac{1}{4}x + \frac{25}{4}$
- Q6.** Lines $y = 2x + 1$ and $y = mx + 3$ are perpendicular. What is the value of m ?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) -2
(D) 2
(E) 0
- Q7.** Given two lines with slopes m_1 and m_2 , what condition must be met for the lines to be perpendicular?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = -1$
(D) $m_1 \cdot m_2 = 1$
(E) $m_1 + m_2 = 0$
- Q8.** If two lines intersect at a 90° angle, what is the relationship between their slopes m_1 and m_2 ?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = -1$
(D) $m_1 \cdot m_2 = 1$
(E) $m_1 + m_2 = 0$

Open Ended Questions (FRQ)

- Q9.** Find the equation of the line passing through $(2, 3)$ that is perpendicular to $y = 5x - 2$. Show all work.
- Q10.** Given two lines $y = 2x + 1$ and $y = -\frac{1}{2}x + 3$, find the point of intersection and verify that the lines are perpendicular. Show all work.

Chef's Tips

- Tip 1: Ensure students understand the concept of negative reciprocal slopes for perpendicular lines.
- Tip 2: Have students practice applying the concepts to real-world problems.
- Tip 3: Encourage students to show their work and verify their answers for free-response questions.